

IT351 - Internet of Things
IoT Sensor Data Prediction Dashboard Using Machine Learning

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1. Introduction

In this project we built an IoT sensor data dashboard that uses machine learning to predict temperature values. IoT devices are commonly used in smart homes, factories, and environmental monitoring. These devices produce a lot of sensor data that needs to be analyzed. Our project covers preprocessing, EDA, a Random Forest prediction model, and a Streamlit dashboard. The sensors we worked with are temperature, humidity, AQI, and CO₂.

2. Dataset

We generated a synthetic IoT dataset because we could not find a free dataset with all the sensors we needed. It covers 60 days with hourly readings (1,440 data points per sensor). We added 3% random missing values to simulate real sensor dropout.

Sensor	Unit	Description	Range
Temperature	deg C	Daily sinusoidal cycle, peaks around 2pm	18 - 37
Humidity	%	Inversely related to temperature	20 - 100
AQI	-	Higher during morning and evening rush hours	0 - 200
CO ₂	ppm	Higher during peak activity hours	380 - 700

3. Data Preprocessing

Missing Values: We used forward fill (ffill) to fill in missing readings because in time series the previous reading is usually close to the missing one. bfill was used for any missing values at the start. **Outlier**

Removal: We used the IQR method and clipped values outside $Q1-1.5*IQR$ and $Q3+1.5*IQR$. **Feature**

Engineering: We extracted hour, day of week, day of year, month, and a weekend flag from the timestamp. We also added lag features at t-1, t-2, and t-24 hours which improved the model significantly.

4. Exploratory Data Analysis

Figure 1 shows the sensor readings for the first 7 days. The daily temperature cycle is clearly visible, and humidity moves in the opposite direction as expected.

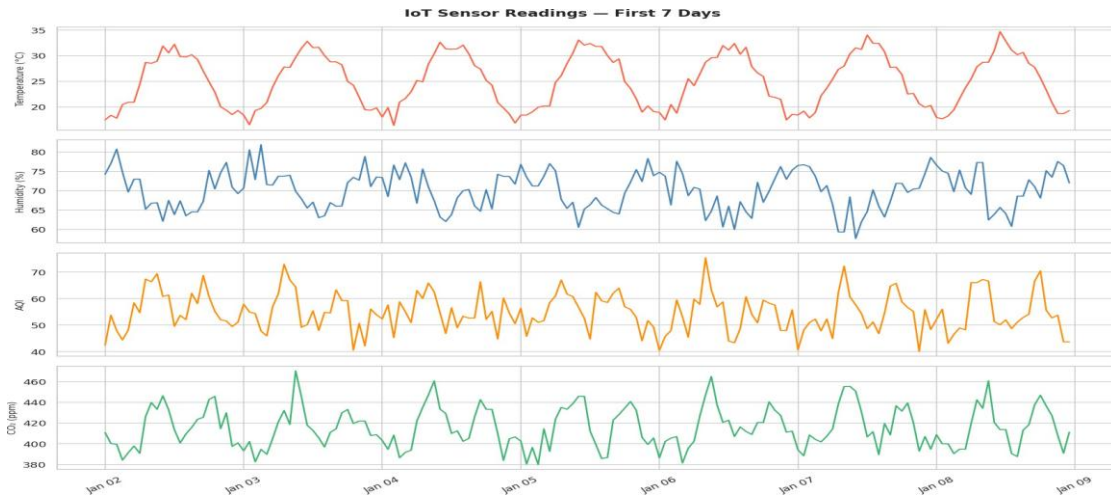


Figure 1: Sensor readings for the first 7 days

Figure 2 shows hourly averages. Rush hour peaks in AQI and CO₂ at 8am and 5pm are visible.

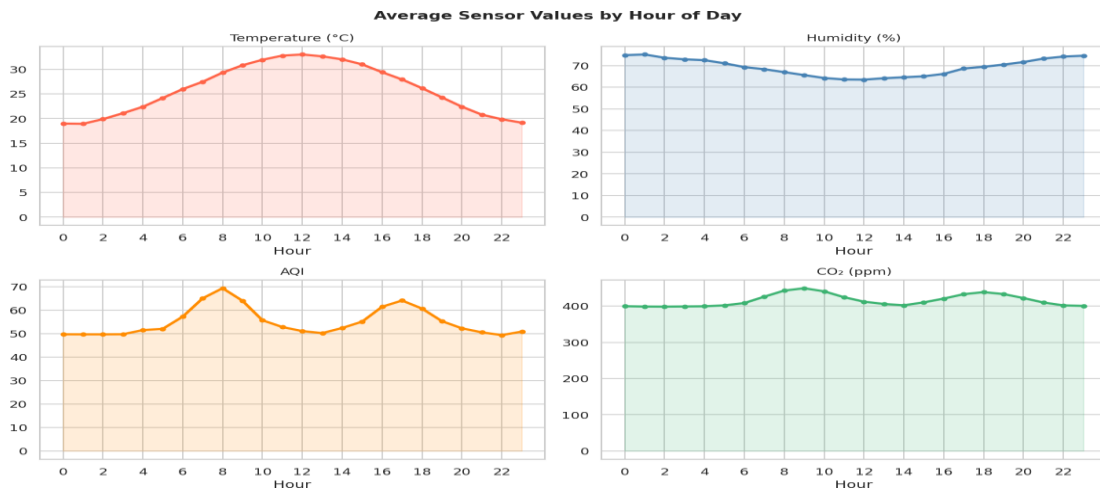


Figure 2: Average sensor values by hour of day

Figure 3 is the correlation heatmap. Temperature and humidity have a strong negative correlation (-0.87) because hot air holds more moisture. AQI and CO₂ are positively correlated (0.55) because both go up with traffic activity.

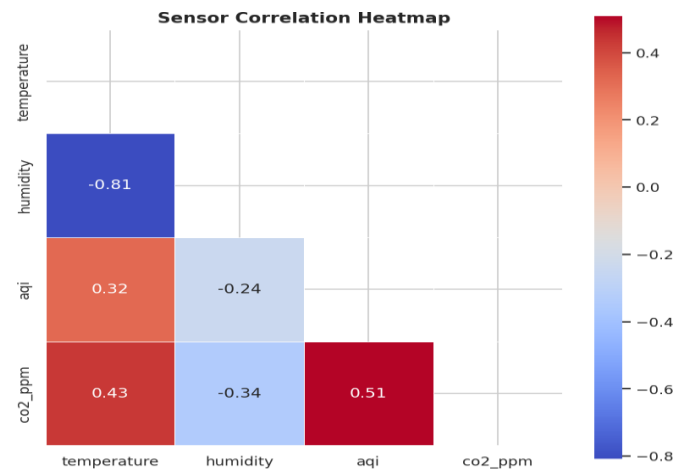


Figure 3: Correlation heatmap between sensors

Figure 4 shows that temperature and humidity are roughly normally distributed. AQI and CO₂ are slightly skewed right because of activity-hour peaks.

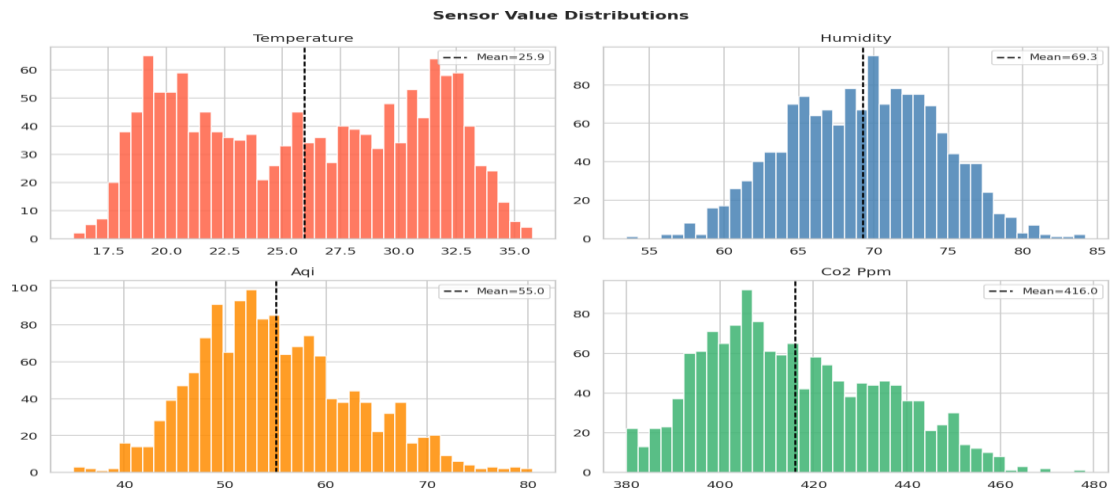


Figure 4: Distribution of sensor values

5. Machine Learning Model

We predicted temperature using 11 features. We did a chronological 80/20 train-test split instead of random split because random split would let the model see future data. Training: 1,152 samples. Test: 288 samples.

Random Forest ($n_estimators=150$, $max_depth=12$) was our main model. We chose it because it handles non-linear patterns well. **Linear Regression** was used as a baseline.

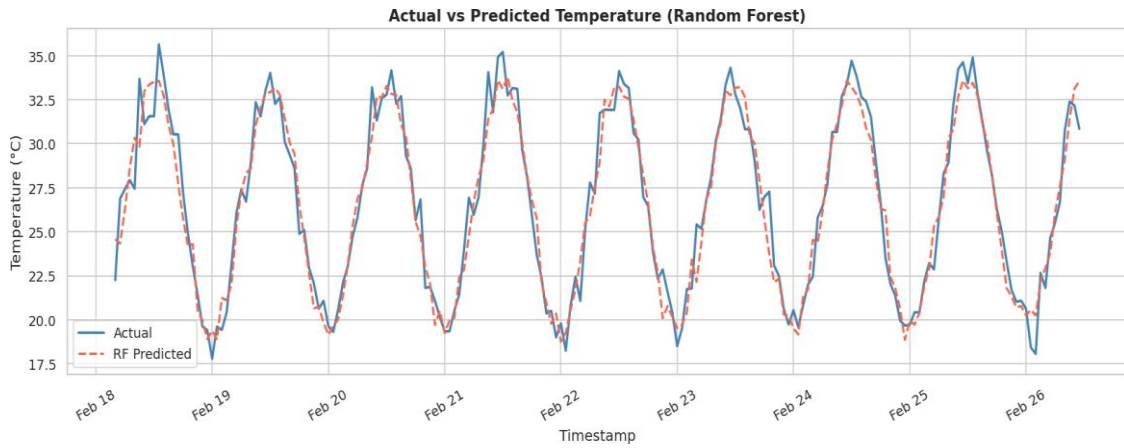


Figure 5: Actual vs predicted temperature on the test set

Figure 6 shows feature importances. The lag features (`temp_lag1`, `temp_lag24`) are the most important which makes sense because temperature does not change suddenly.

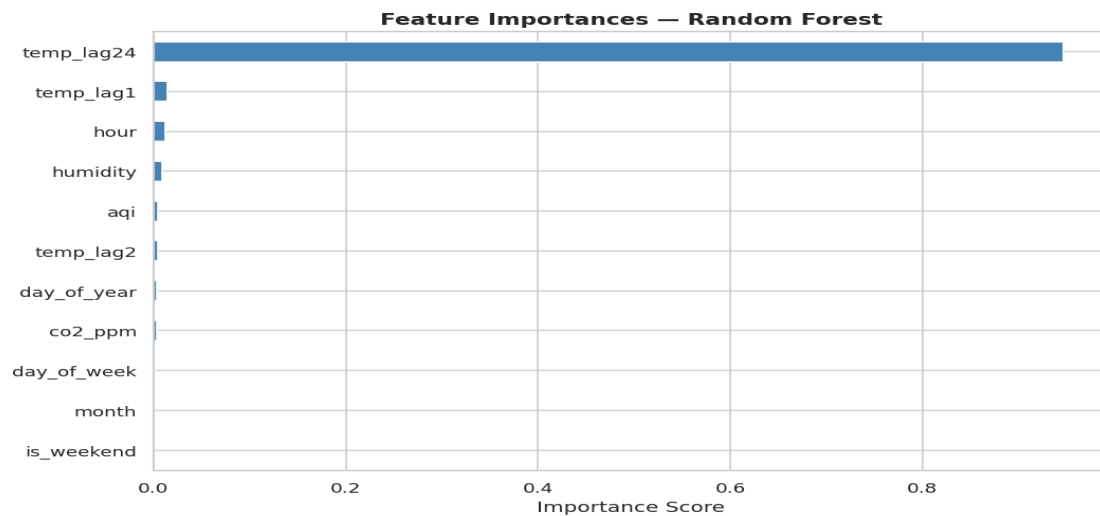


Figure 6: Random Forest feature importances

6. Model Evaluation

We used MAE, RMSE, and R2 to compare the models:

Metric	Random Forest	Linear Regression
MAE (deg C)	0.9699	1.0345
RMSE (deg C)	1.2275	1.2904
R2 Score	0.9402	0.9339

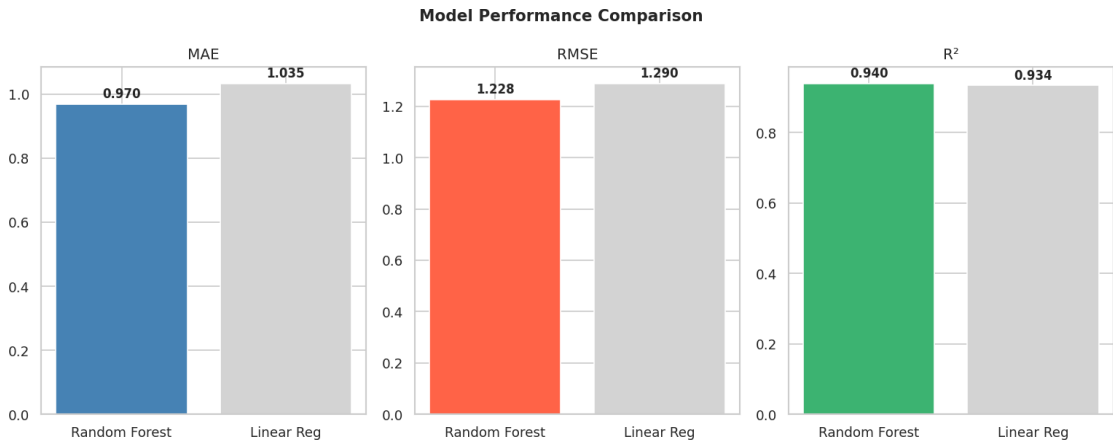


Figure 7: Model performance comparison

Random Forest got $R^2 = 0.9402$ which means it explains about 94% of the variance in temperature. The MAE is 0.97 degrees which is quite good for hourly predictions. Linear Regression also performed well ($R^2 = 0.9339$) but Random Forest is clearly better.

7. Dashboard

We built the dashboard using Streamlit. It has 4 pages accessible from the sidebar, a slider to control how many days of data to display, and a dropdown to select the primary sensor.

Page	Content
Overview	KPI metric cards, sensor distributions, correlation heatmap
Trends	Time series and hourly average charts for selected sensor
ML Predictions	Actual vs predicted chart, feature importance, residual analysis
Model Metrics	MAE, RMSE, R2 comparison table and bar charts for both models

To run the dashboard:

```
pip install streamlit scikit-learn pandas numpy matplotlib seaborn
streamlit run dashboard.py
```

8. Conclusion

In this project we built a complete IoT sensor analysis and prediction system. The Random Forest model achieved $R^2 = 0.9402$ and $MAE = 0.97$ degrees which we are happy with. The most important finding was that lag features are very powerful for time series prediction. The Streamlit dashboard makes the results easy to understand and explore.

In the future we would like to use real sensor data from Kaggle, add anomaly detection, and try LSTM networks which are designed for time series data.

9. References

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